CMSC433 / Part 06

Verification conditions

Guided practice

Use with the chat: do Question 1 first; then derive one loop obligation at a time. These are partial-correctness questions; omit termination obligations.

Worked example. For `x := x - 2;` with postcondition $x \geq 4$, substitution gives $x-2 \geq 4$, equivalently $x \geq 6$. Work backward in a sequence.

1. Weakest precondition

Compute the weakest precondition of `x := y + 2; y := x - 1;` for postcondition $x > y \wedge y \geq 0$. Show the intermediate formula.

Answer: Before last assignment: [ ]; before first: [ ].

2. One invariant, three obligations

```
method Twice(n:int) returns (t:int)
  requires n >= 0
  ensures t == 2*n + 1
{
  t := 0;
  var i := n;
  while i > 0 {
    t := t + 2;
    i := i - 1;
  }
  t := t + 1;
}
```

Use invariant $I \equiv i \geq 0 \wedge t + 2i = 2n$. Derive initialization, preservation, and exit, substituting the code into each. Account for the final assignment when deriving the exit obligation.

Answer: Initialization: [ ].

Answer: Preservation: [ ].

Answer: Exit: [ ].

Source: Sample Q3, p. 3; DafnyVerificationConditions physical pp. 3–30.